MODULE: HDIP in Sci in Data Analytics for Business *Machine Learning for Business*

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https://github.com/Danti3341/-Integrated-CA2--Machine-Learning-for-Business-HDip-in-Data-Analytics

INTRODUCTION

The report presents an analytical overview of a series of visualisations designed to explore customer behaviour, satisfaction, and demographic patterns within a fashion rental service. Each chart provides a different perspective on user interactions, ranging from numerical ratings and written sentiment to body characteristics, fit feedback, and rental preferences. Collectively, the visualisations offer a comprehensive understanding of how customers evaluate their experiences, how their physical attributes influence rental patterns, and how inventory is structured to meet demand. The summaries highlight several consistent findings: most customers express highly positive satisfaction levels, both quantitatively through ratings and qualitatively in written reviews; rental activity centres around a small number of popular clothing categories; and demographic patterns such as age, height, weight, and body type show predictable yet diverse distributions. The various chart types, bar charts, strip plots, line charts, tree maps, scatterplots, boxplots, and histograms were chosen for their ability to clearly communicate trends, proportions, variability, and relationships within the dataset. Together, they provide an integrated and accessible portrayal of the customer base and their interactions with the service.

Chart 1: Distribution of Customers' Ratings

The bar chart shows that most customers gave very high ratings, with rating 5 being the most common, followed by rating 4. Ratings 3, 2, and 1 appear much less often, indicating that negative experiences were rare and overall customer satisfaction was high. The values displayed confirm this pattern: ratings 4 and 5 dominate the distribution, while lower ratings represent only a small portion of responses. The bar chart is an appropriate choice because it clearly compares the frequency of each rating category. Different colours are used to distinguish the rating levels, making the visual easy to interpret and helping highlight the strong trend toward positive evaluations.

Chart 2: Distribution of Ratings by Occasion

The chart shows that customer ratings remain consistently high across all occasions, with most ratings clustering at 4 and 5. This indicates that users are generally satisfied regardless of whether the product is used for casual, social, or formal events. Lower ratings appear only rarely and are evenly distributed, suggesting that no specific occasion is linked to poor experiences. The values displayed confirm this trend, as high ratings dominate categories such as weddings, dates, and formal affairs. The strip plot is appropriate because it shows individual data points and allows clear comparison across occasions. The use of different colours for each occasion improves readability and helps distinguish rating patterns across contexts.

Chart 3: Average Trend Rating Over Time

The chart shows how the average customer rating changed from 2011 to 2018. At the beginning, ratings fluctuate sharply, likely due to limited early data or operational adjustments. From mid-2012 onward, the trend becomes stable, with average ratings consistently staying between 4.4 and 4.6. This stability indicates long-term customer satisfaction and reliable product performance. The values displayed confirm this pattern, as the early variation gradually narrows into a consistent range of positive ratings. A line chart is appropriate for showing changes over time, and the use of a soft green line ensures clarity while highlighting the continuous nature of the trend without distracting from the data.

Chart 4: Top 20 Most Popular Categories

The chart shows the top 20 most rented clothing categories, with “dress” being by far the most popular, followed by “sheath,” though at less than half the volume. Other categories, such as jumpsuits, maxi dresses, jackets, and skirts, have moderate rental numbers, while the least rented items—like down jackets, vests, and shawls—show very low demand. This pattern indicates that customer interest is heavily concentrated in a few key categories, especially dresses. The values displayed highlight a steep decline from the most to the least popular items, illustrating uneven customer preferences. A horizontal bar chart is used because it clearly compares categories with long labels. The blue colour gradient helps show differences in rental volume, with darker shades signalling higher demand and lighter shades representing lower frequencies.

Chart 5: Categories Within Occasions

The chart shows the top 20 most rented clothing categories, with “dress” being by far the most popular, followed by “sheath,” though at less than half the volume. Other categories, such as jumpsuits, maxi dresses, jackets, and skirts, have moderate rental numbers, while the least rented items—like down jackets, vests, and shawls—show very low demand. This pattern indicates that customer interest is heavily concentrated in a few key categories, especially dresses. The values displayed highlight a steep decline from the most to the least popular items, illustrating uneven customer preferences. A horizontal bar chart is used because it clearly compares categories with long labels. The blue colour gradient helps show differences in rental volume, with darker shades signalling higher demand and lighter shades representing lower frequencies.

Chart 6: Age Distribution by Body Type

The chart shows how customer ages are distributed across different body types, with all groups following a similar pattern. Most customers fall between the ages of 25 and 40, indicating that the service is mainly used by adults in this age range. Very young and very old users appear only rarely. While most body types share similar shapes and concentrations, categories like petite, apple, and full bust show slightly more variation. The violin plot displays the full age distribution for each group, with wider areas representing more common ages. Scatter points help illustrate individual observations. This chart type is appropriate because it shows both density and variability, and the use of soft, distinct colours helps differentiate the body-type categories clearly.

Chart 7: Height vs. Weight Correlation

The scatterplot shows a positive relationship between height and weight, meaning that taller customers tend to weigh more. Most users fall within typical adult ranges—100 to 200 lbs and 60 to 70 inches—although a few individuals lie outside these values. The points demonstrate considerable variation, showing that body types cover a wide range of measurements rather than specific height–weight combinations. Each point represents one customer, and colour coding highlights how body types are distributed across the chart. The scatterplot is an appropriate choice because it effectively displays correlations, variability, and outliers. The use of different colours helps distinguish body types and makes patterns easier to observe.

Chart 8: Distribution of User Body Types

The tree map shows the proportion of users belonging to different body-type categories. The largest groups are hourglass, athletic, and petite, while pear and straight & narrow also represent a meaningful share. Smaller groups, such as full bust, apple, and unknown, make up only a small part of the user base. This uneven distribution suggests that certain body types may generate higher demand for specific garments. Each block in the tree map represents a body type, and its size reflects how many users belong to that category. The chart is effective because tree maps clearly show proportions through area size. The use of different colours for each body type helps distinguish the categories and makes the visual easy to interpret.

Chart 9: Fit Feedback Breakdown by Body Type

The chart compares fit feedback across different body types and shows that hourglass, athletic, and pear users provide the most responses. Across all groups, most customers report that items fit well, while fewer indicate that items are too large or too small. Larger body-type groups naturally contribute more feedback overall. The values displayed in the stacked bars show both the total amount of feedback and the proportion of each fit category, with green indicating proper fit and smaller orange and blue sections representing sizing issues. The stacked bar chart is appropriate because it clearly compares fit outcomes across categories, and the colour scheme helps differentiate between fit, large, and small feedback, making the patterns easy to interpret.

Chart 10: Size Range Distribution by Top 20 Categories

The chart shows how garment sizes vary across the top 20 rented clothing categories. Some categories—such as gowns, sheath dresses, maxi dresses, and shirtdresses—have a wide size range, indicating they are offered in many sizes to meet diverse customer needs. Others, like tops, pants, sweaters, and rompers, have narrower ranges, suggesting more standardised sizing or less variation in demand. The values displayed in each boxplot show the median size, the spread of the middle 50% of sizes, and any outliers. This helps compare how much size variation exists across categories. The boxplot is an appropriate chart type because it clearly summarises size distribution and variability. Different colours are used to distinguish each category, while scatter points add detail by showing individual size values.

Chart 11: Sentiment Analysis of Review Texts

The chart shows the distribution of sentiment scores from customer reviews, revealing that most comments are positive. Sentiment values between 0.1 and 0.6 are the most common, while neutral or slightly negative reviews are less frequent, and strongly negative reviews are rare. This pattern supports the conclusion that customers generally express favourable experiences. The histogram bars display how many reviews fall into each sentiment range, with green representing positive sentiment and red representing negative. The predominance of green bars highlights the strong positive trend. A histogram is appropriate because it clearly shows the frequency of different sentiment values, and the colour scheme helps distinguish positive from negative feedback, making the emotional tone easy to interpret.

Chart 12: Tree map of Positive and Negative Feelings by Rating

The tree map shows how positive, neutral, and negative emotions are distributed across different rating values. Positive sentiment dominates the dataset, with ratings of 5.0 and 4.0 occupying the largest areas, indicating that these high ratings contribute most to favourable emotional responses. Neutral and negative sentiments appear much less frequently, and strongly negative feelings are rare. Each rectangle's size represents how often a rating occurs within each sentiment category, making it easy to see that higher ratings are strongly linked to positive emotions. The tree map is an effective chart type because it visually represents proportions within hierarchical categories. The colour scheme—green for positive, grey for neutral, and red for negative—helps the viewer quickly distinguish emotional categories and understand the overall sentiment pattern.

Conclusion

The visualisations collectively show that customers are highly satisfied with the service, as reflected in the consistently high ratings and predominance of positive sentiment. The charts also reveal clear patterns in customer behaviour, including strong preferences for certain clothing categories and predictable demographic trends. Fit feedback, size distributions, and body-type data highlight that most users find items appropriate for their needs. Overall, the graphs demonstrate that the dataset is rich, well-structured, and highly suitable for machine learning applications, providing valuable insights that can support better recommendations, improved inventory management, and enhanced customer experience.

Finally, the choice of chart types and colour usage strengthens the clarity and interpretability of all visualisations. Bar charts and stacked bars effectively compare counts across categories, line charts show trends over time, scatterplots reveal correlations, and boxplots summarise distributions. Tree maps communicate proportional relationships between categories, while histograms display sentiment distributions. The colour schemes are intentionally chosen to distinguish categories clearly, such as green for positive sentiment, red for negative, and varied palettes for body types and garment categories, making comparisons intuitive and visually accessible. Together, these graphical tools and colour strategies ensure that complex datasets are presented in a clear, academically coherent, and easy-to-interpret manner.

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